Random Forests and Other Learner Ensembles

From Single Trees to Forest Wisdom

Random Forests and Other Learner Ensembles

A Couple of Notable Criteria for A Good Prediction Model

Discuss bias and variance tradeoffs and interpretability/prediction performance tradeoffs.

Choosing the Right Bias-Variance Tradeoff

The bias-variance tradeoff is fundamental to understanding model performance. As model complexity increases, bias decreases but variance increases, creating an optimal point where total error is minimized.

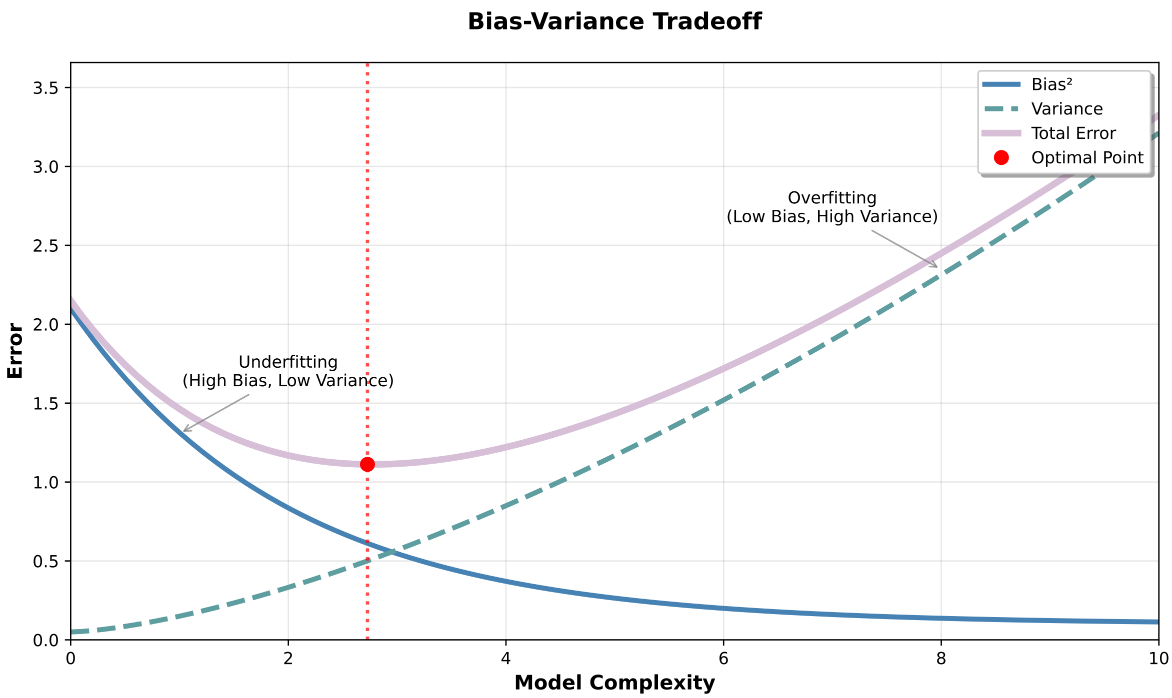
> ℹ What Are Bias and Variance?
>
> **Bias** (from statistics) is the difference between the expected (average) prediction of your model and the true value. It measures systematic deviation - how far off your model is on average.
> **Variance** (from statistics) is how much your model's predictions vary when trained on different datasets. It measures instability - how much your predictions scatter around their average.
> **The etymology:** - **Bias** comes from archery - it means the arrow consistently misses the target in the same direction - **Variance** comes from statistics - it measures how spread out values are from their mean
> **Mathematically:** - **Bias²** = (Average Prediction - True Value)² - **Variance** = Average of (Individual Prediction - Average Prediction)²
> **Important clarification:** The variance in bias-variance decomposition measures how much your *model's predictions* vary across different training datasets - it's independent of the true values. This is different from the inherent variability in the true values themselves.

Practical measurement: Since we rarely have truly "different" datasets, we estimate variance using techniques like bootstrap sampling, cross-validation, or bagging (like Random Forests do). Don't worry if these terms are unfamiliar - we'll explore them as we learn about ensemble methods.

Connection to precision and accuracy: - **Bias** is essentially **inaccuracy** - low bias means your model is accurate (close to truth on average) - **Variance** is essentially **imprecision** - low variance means your model is precise (consistent predictions)

Key difference: Precision/accuracy are typically measured on a single model's predictions, while bias/variance are theoretical concepts about how a model would behave across many different training datasets.

Simple analogy: - **High bias, low variance**: A broken clock (always wrong, but consistently wrong) = inaccurate but precise - **Low bias, high variance**: A drunk person trying to hit a target (might be right on average, but wildly inconsistent) = accurate on average but imprecise



This graph illustrates the fundamental tradeoff in machine learning: simple models have high bias but low variance (underfitting), while complex models have low bias but high variance (overfitting). The optimal model complexity occurs where the total error is minimized, balancing both sources of error.

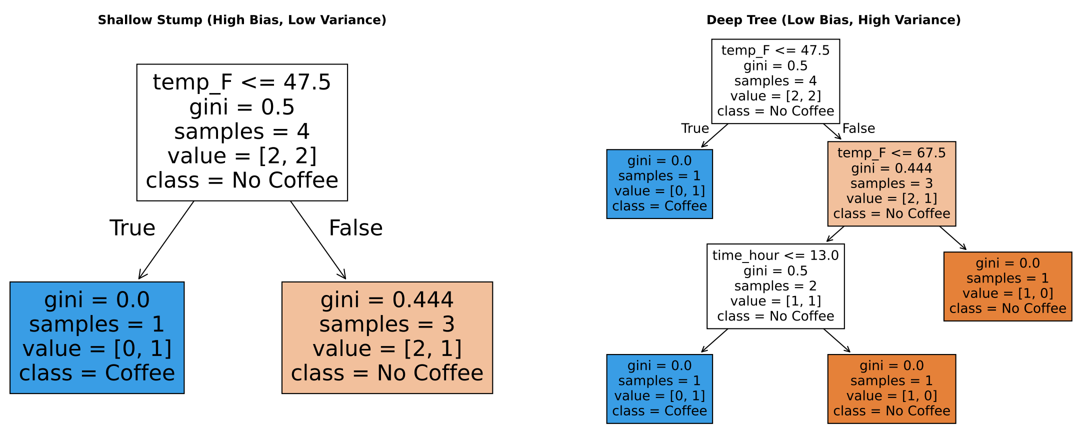
Examples of Bias and Variance

Let's explore bias and variance with a concrete example: predicting whether someone buys hot coffee based on time of day, temperature, and rain. People usually buy hot coffee in the morning, but extreme heat suppresses it. In the evening, most skip it unless it is very cold and raining.

```
Coffee Dataset:
   time_hour  temp_F  rain  bought_hot_coffee
0          8      55     0                  1
1          8      80     0                  0
2         18      55     0                  0
3         18      40     1                  1
```

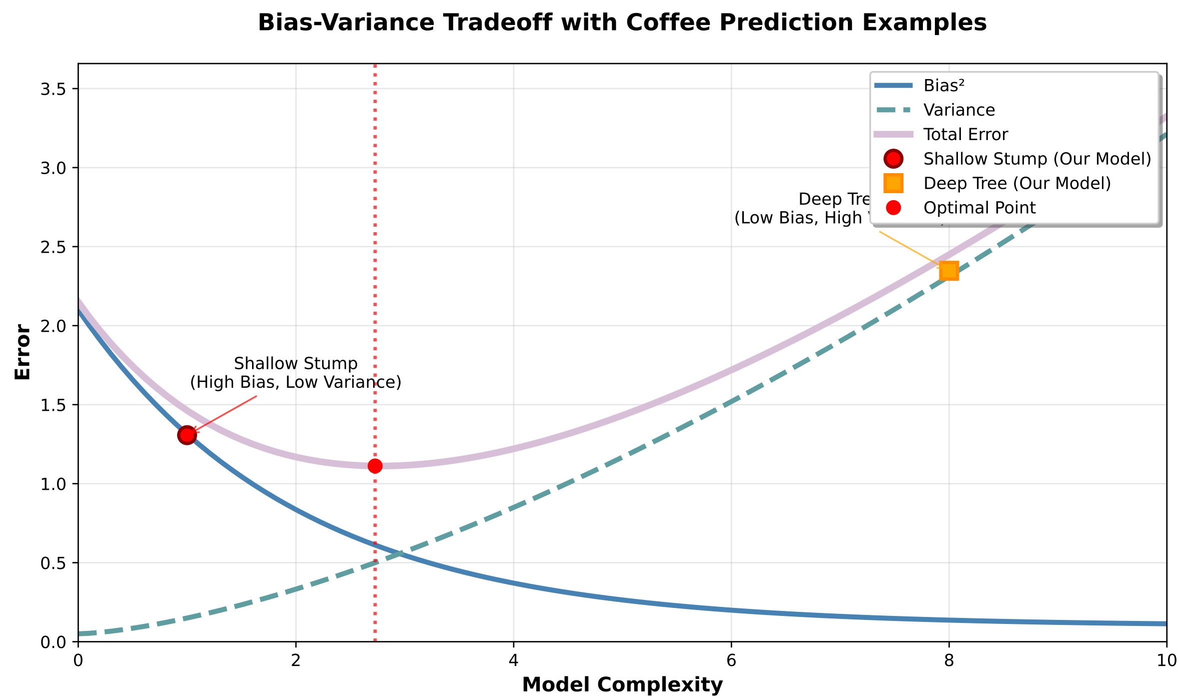
```
Pattern: Morning coffee (8am) unless too hot (80°F), evening coffee (6pm) only when cold and
```

Now let's fit two models to demonstrate the bias-variance tradeoff:




```
Predictions:
Shallow stump predictions: [0 0 0 1]
Deep tree predictions: [1 0 0 1]
Actual values: [1 0 0 1]
```

The shallow stump makes a simple rule but misses the nuanced pattern (high bias). The deep tree memorizes the training data perfectly but would likely fail on new data (high variance).



This example shows how the shallow stump (high bias) makes simple but inaccurate rules, while the deep tree (high variance) memorizes the training data but would likely perform poorly on new coffee-buying scenarios. The optimal model would balance both sources of error.

Choosing the Right Interpretability/Prediction Performance Tradeoff

The Options In Light of These Criteria

A single weak learner

A strong learner

An ensemble of weak learners

An ensemble of strong learners

The Core Idea

While individual decision trees provide interpretable, rule-based predictions, they often suffer from overfitting and instability. Random forests and other ensemble methods address these limitations by combining multiple learners to create more robust and accurate predictions. These ensemble approaches represent a fundamental evolution from single-model thinking to collective intelligence in machine learning.

Single Decision Trees

Provide interpretable rules but can overfit and be unstable to small data changes.

Example: One tree might predict anxiety based on stress > 6, but this rule could change dramatically with new data.

Random Forests

Combine hundreds of trees, each trained on different data samples and feature subsets.

Example: The forest averages predictions from 100 trees, each seeing slightly different data, creating more stable and accurate anxiety predictions.

The Ensemble Advantage

The power of ensemble methods lies in the principle that multiple weak learners, when combined thoughtfully, can outperform any single strong learner. This concept, known as the "wisdom of crowds" in machine learning, addresses the fundamental trade-offs between bias and variance that plague individual models.